

Snowflake IAM Role & S3 Integration – End-to-End Explanation

This document provides an end-to-end explanation of how Snowflake securely accesses Amazon S3 using an IAM Role. It explains why an AWS Account is selected as the trusted entity, how the External ID is used, and how to validate the configuration from Snowflake.

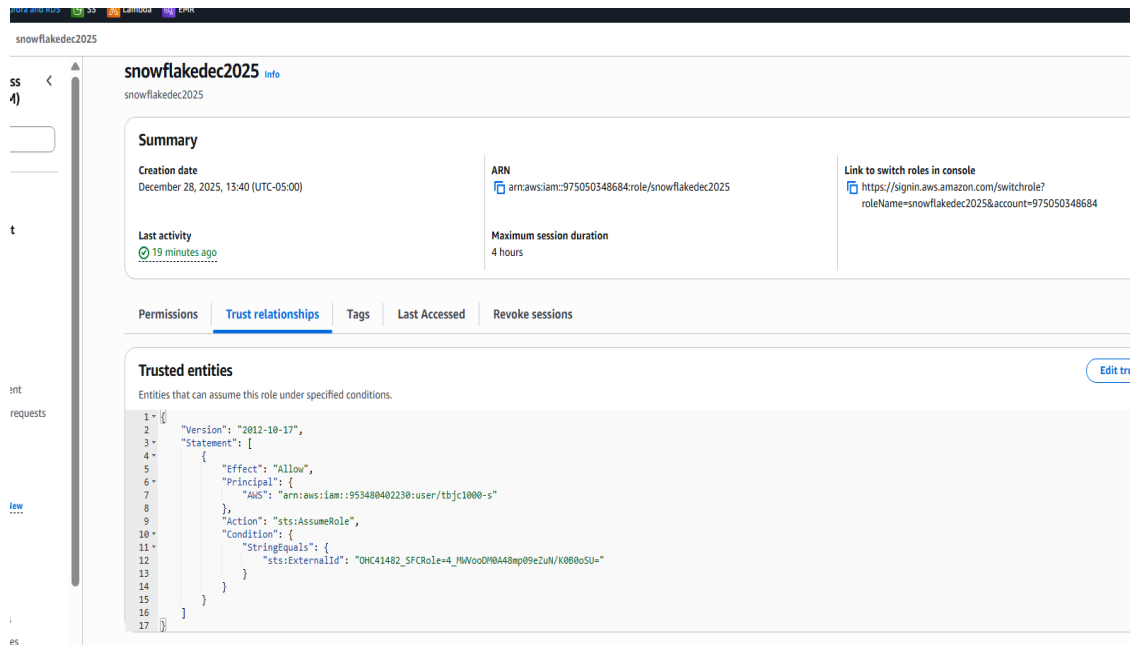
1. Selecting AWS Account as Trusted Entity

During IAM role creation, the trusted entity is set to **AWS account** (not AWS service). This is required because Snowflake is a third-party SaaS application and not an AWS-managed service. Snowflake assumes the role using cross-account STS AssumeRole calls.

The screenshot shows the AWS IAM console 'Create role' page, specifically the 'Select trusted entity' step. The 'Trusted entity type' section has four options: 'AWS service' (selected), 'AWS account' (selected), 'Web identity', and 'SAML 2.0 federation'. The 'AWS account' option is highlighted with a blue border. Below this, the 'An AWS account' section shows 'This account (975050348684)' selected. The 'Options' section has 'Require external ID (Best practice when a third party will assume this role)' checked. The 'External ID' field is empty, and the 'Require MFA' checkbox is unchecked. A blue 'Cancel' button is visible at the bottom right.

The **Require external ID** option is enabled to protect against the confused deputy problem. Snowflake later provides this External ID as part of the storage integration metadata.

2. IAM Role Trust Policy (Snowflake Principal)



The screenshot shows the AWS IAM console for the role **snowflakedec2025**. The **Trust relationships** tab is active, showing the trust policy JSON. The policy allows the principal `arn:aws:iam::953480402230:user/tbjc1000-s` to perform the `sts:AssumeRole` action, with a condition that the `sts:ExternalId` must equal `"0HK41482_SFRole=4_YmVooDf0B4A8mp09eZuH/K0B0o5U="`.

Summary

- Creation date:** December 28, 2025, 13:40 (UTC-05:00)
- ARN:** `arn:aws:iam::975050348684:role/snowflakedec2025`
- Link to switch roles in console:** <https://signin.aws.amazon.com/switchrole?roleName=snowflakedec2025&account=975050348684>
- Last activity:** 19 minutes ago
- Maximum session duration:** 4 hours

Trusted entities

Entities that can assume this role under specified conditions.

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Principal": {
7         "AWS": "arn:aws:iam::953480402230:user/tbjc1000-s"
8       },
9       "Action": "sts:AssumeRole",
10      "Condition": {
11        "StringEquals": {
12          "sts:ExternalId": "0HK41482_SFRole=4_YmVooDf0B4A8mp09eZuH/K0B0o5U="
13        }
14      }
15    }
16  ]
17 }
```

The trust policy explicitly allows a Snowflake-owned AWS IAM user to assume the role using `sts:AssumeRole` along with the External ID. This establishes secure cross-account access.

3. Snowflake Storage Integration – External ID

snowflakedec2025

Info

snowflakedec2025

Summary

Creation date

December 28, 2025, 13:40 (UTC-05:00)

ARN

arn:aws:iam::975050348684:role/snowflakedec2025

Link to switch roles in console

<https://signin.aws.amazon.com/switchrole?roleName=snowflakedec2025&account=975050348684>

Last activity

18 minutes ago

Maximum session duration

4 hours

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (1)

Info

You can attach up to 10 managed policies.

Filter by Type

All types

Search

< 1 >

☐	Policy name	Type	Attached entities
☐	AmazonS3FullAccess	AWS managed	18

▶ Permissions boundary (not set)

▼ Generate policy based on CloudTrail events

You can generate a new policy based on the access activity for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

App

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Snowflake generates a unique External ID and IAM User ARN as part of the storage integration. These values must match the IAM role trust policy for successful access.

4. DESC INTEGRATION Verification

My Workspace > awssnowpipes.sql

ACCOUNTADMIN SAMPLE_NEWWH (Small) MYDB PUBLIC

```
1  -- AWS S3 Bucket
2
3  create or replace storage integration s3_int12
4  TYPE = EXTERNAL_STAGE
5  STORAGE_PROVIDER = S3
6  ENABLED = TRUE
7  STORAGE_AWS_ROLE_ARN = 'arn:aws:iam::975050348684:role/snowflakedec2025'
8  STORAGE_ALLOWED_LOCATIONS = ('s3://snwoflakedec2025/csv/')
9  COMMENT = 'integration object' ;
10
11  DESC integration s3_int12;
```

Results (just now)

Table Chart 9 rows 92ms

	property	property_type	property_value
1	ENABLED	Boolean	true
2	STORAGE_PROVIDER	String	S3
3	STORAGE_ALLOWED_LOCATIONS	List	s3://snwoflakedec2025/csv/
4	STORAGE_BLOCKED_LOCATIONS	List	
5	STORAGE_AWS_IAM_USER_ARN	String	arn:aws:iam::953480402230:user/tbjc1000-s
6	STORAGE_AWS_ROLE_ARN	String	arn:aws:iam::975050348684:role/snowflakedec2025
7	STORAGE_AWS_EXTERNAL_ID	String	OHC41482_SFCRole=4_MWVooDM0A48mp09eZuN/K0B0oSU=
8	USE_PRIVATELINK_ENDPOINT	Boolean	false
9	COMMENT	String	'integration object'

The DESC INTEGRATION output confirms that the Snowflake principal, role ARN, and External ID are correctly configured. Matching values indicate a successful Snowflake–S3 integration.